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# How Constructive Design Researchers Drift: Four Epistemologies

Peter Gall Krogh, Ilpo Koskinen

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**Introduction**

Constructive design research is a form of research in which design itself becomes an act of producing knowledge. For example, Joep Frens claimed that a camera without buttons and menus would lead to a better user experience than a conventional camera does.[1] To examine the claim, he built a camera with four interface variations to test his hypothesis with people. With the Presence Project, William Gaver and colleagues built imaginative interaction devices to examine whether they could bring communities together.[2] Andrea and Marcelo Judice created graphics to help doctors and nurses to manage multi-resistant tuberculosis.[3] Each of these studies advanced knowledge of interaction and the latter how to design social forms that can heal people. What is common to them is that they could not have been done without design, either theoretically or historically.

In less than 20 years, this form of research has become mainstream in design universities; its methodology has been described elsewhere.[4] Less attention has been paid to the epistemological background of this form of research: how it produces knowledge and what drives knowledge production in it. This article examines this question by describing four epistemological traditions at work in constructive design research.

The way in which we approach epistemology is through the concept of drifting.[5] We define drifting as engaging in those informed actions that take design away from its original brief or question and that lead to a result that was not anticipated in the beginning. We argue that drifting is an intentional, methodic process. Designers do research, create concepts, test these concepts with sketches and mock-ups, and evaluate them repeatedly to go beyond the brief in their quest for a preferable outcome. We also argue that drifting is a function of epistemological beliefs—beliefs about knowledge. Our earlier work has identified several types of drift.[6] One designer believes that the studio is the site of knowledge: Her design changes direction when she discovers a better way to bend wood in the studio. Another locates knowledge in the design community and changes his direction when his fellow designers critique him. A third designer drifts when her simulations show

1 Joep Frens, "Designing for Rich Interaction: Integrating Form, Interaction, and Function," Eindhove, Technische Universiteit Eindhoven, 2006.

2 *The Presence Project*, CRD Projects Series (London: Royal College of Art, 2001).

3 Andrea Judice, "Design for Hope," (Phd diss., Aalto University, 2014).

4 See, e.g., Ilpo Koskinen et al., *Design Research Through Practice: From the Lab, Field, and Showroom* (Waltham, MA: Morgan Kaufmann, 2011); Thomas Markussen and Anne Louise Bang, "Dynamic Research Sketching: A New Explanatory Tool for Understanding Theory Construction in Design Research," in *Proceedings of the 8th International Design Research Society Conference* (Thailand: Chulalongkorn University Bangkok, 2012), 1126–43; and Thomas Markussen et al., "On What Grounds? An Intra-Disciplinary Account of Evaluation in Research Through Design" in *Proceedings of the 6th International Association of Societies of Design Research Conference* (Brisbane: Australia, 2015), 1415–29.

5 Peter Krogh and Ilpo Koskinen, *Drifting by Intention: Four Epistemic Traditions from Within Constructive Design Research* (Dordrecht: Springer, 2020).

6 Peter Gall Krogh et al., "Ways of Drifting—Five Methods of Experimentation in Research Through Design" in *Proceedings of ICoRD15—Research into Design Across Boundaries* (New Delhi: Springer, 2015), 39–50.

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that something is wrong with the structure she was about to propose, and a fourth when his artistic intuition tells him so. A fifth is guided by experimental research designs and a sixth by users in a co-design process. Practicing designers know that well-executed design projects can reveal that the problem that launched a project on its way is not always the true problem. Epistemology matters.

We believe that design readers recognize the meaning of the term *drifting*, but it exposes a bias in design research that may not be obvious. Design literature is rich in terms and notions that describe the shift from brief to outcome in radical, innovation-based terms.[7] However, these frameworks have roots in business and have limited currency in design research. A similar tendency to dramatize innovation in research exists in literature that tries to understand theoretical change in design research.[8] In contrast, we believe that a much wider array of shifts exists and must be acknowledged in design literature. The concept of drifting is designed to expose this array of shifts for research. This article takes the first step by explicating some epistemological beliefs in design research and by showing how these beliefs are related to drifting.

**Epistemology from Within**

This article is interested in design research rather than design practice. Specifically, we are interested in how concepts of knowledge interact with drifts in research—that is, how designers' beliefs about what constitutes knowledge justify and invite those changes of direction we call drifting. We do not claim that designers' concept of knowledge—their epistemology, often implicit—is the only force that affects drifting. However, it is important enough to merit attention, and unlike their commercial peers, design researchers need to understand the interaction between epistemology and drifting if they want to justify changes of direction in research. Being aware of epistemology is an important component of quality control for them.

Our analysis of epistemology started from existing epistemological writing in design and engineering, but we found little of it, with one exception.[9] The few sources we managed to find point out that epistemology in the world-making disciplines is heavily embedded in culture and society at large.[10] As Carl Mitcham points out, because everything is human-made, computers are embodiments of human values; and as such, they give a material form to ethical problems that should be studied epistemologically. His analysis is useful and consistent with how we see "design from within," but he does not provide a method for explicating these values.

Our perspective on epistemology follows that of Anne-Françoise Schmid, who seeks to rethink a relationship of philosophy and design in which objects no longer are fixed and epistemology has

7 Roberto Verganti, *Design-Driven Innovation: Changing the Rules of Competition by Radically Innovating What Things Mean* (Harvard Business Press, 2009); and Kees Dorst, *Frame Innovation: Create New Thinking by Design* (Cambridge, MA: MIT Press, 2015).

8 Johan Redström, *Making Design Theory* (Cambridge, MA: MIT Press, 2017).

9 Anne-Françoise Schmid, "The Philosophical Underpinnings of Design Theory," in *Advancements in the Philosophy of Design*, ed. Pieter Vermaas and Stéphane Vial (Dordrecht: Springer, 2018), 415–30.

10 Carl Mitcham, "Computers: From Ethos and Ethics to Mythos and Religion: Notes on the New Frontier Between Computers and Philosophy," *Technology in Society* 8, no 3, (1986): 171–201; and Bas Van Fraassen, *Quantum Mechanics: An Empiricist View* (New York: Oxford University Press, 1991), 79–105.



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collapsed into sociology. Following her view, our perspective builds on historical and empirical studies of science. Studies have examined how science sometimes begins to treat some facts as paradigmatic and how these untouchable truths sometimes are revolutionized.[11] Studies also consider how facts are manufactured in laboratories and research is instituted with ethnographic tools.[12] Many of these arguments have followers in design communities as well—all the way from the implicit neo-platonism of the Bauhaus to Johan Redström's recent Kuhnian-inspired interpretation of design.[13]

In contrast to the traditional approaches to epistemology in philosophy—where epistemology means the study of knowledge and those conditions under which we can trust some statement and take it to be true—laboratory studies pushed us to study epistemology empirically.[14] We have studied some key texts in the literature to explicate their epistemology, which is often implicit. An epistemology like this, written from within, has several practical implications. It tells researchers how to treat previous research and how to select methods and methodologies. It tells them the nuts and bolts of research: which arguments are strong enough to change the direction of research; how to experiment; how to evaluate their findings; and how to communicate the outcomes. It also provides a tool that reveals implicit assumptions in disciplines of design and clarifies the relationship between some of the debates that have shaped, first, design research and then design itself over the past two and a half decades.

Our method has been explication, rather than historical or ethnographic methodologies, and our key construct is "tradition." Merriam-Webster defines tradition as "an inherited, established, or customary pattern of thought, action, or behavior." In the context of epistemology, it refers to habitual, inherited beliefs regarding knowledge that shape research by giving it a basis from which to justify drifting.[15] In choosing this word, we suggest that some concepts of knowledge and knowledge production are so deeply ingrained that they function as seen and unquestioned background in research. Under normal circumstances, they are not questioned or contested. They keep shaping the ways in which researchers perceive and undertake their craft, and they also shape the ways in which researchers interpret the views of the opposing camps.

**Explicating a Corpus**

This article builds on a corpus of PhD-level design research collected systematically over the years from several Northern European countries. Most of the work comes from the border of product design and interaction design; it often has a base in human-computer interaction (HCI), and sometimes it ventures into textile

11 Anne-Françoise Schmid, "The Philosophical Underpinnings of Design Theory."

12 See, e.g., Bruno Latour and Steve Woolgar, *Laboratory Life: The Construction of Scientific Facts* (Princeton: Princeton University Press, 2013); Michael Lynch, *Scientific Practice and Ordinary Action: Ethnomethodology and Social Studies of Science* (Cambridge, UK: Cambridge University Press, 1993); and Harry M. Collins, *Changing Order: Replication and Induction in Scientific Practice* (Beverly Hills, CA: Sage, 1985), 1–28.

13 Otl Aicher, "The Bauhaus and Ulm," in Philipp Oswalt (Ed.), *Bauhaus Conflicts 1919–2009: Controversies and Counterparts* (Ostfildern: Hatje Cantz, 2009), 174–86.

14 Richard Feldman, *Epistemology* (Upper Saddle River, NJ: Prentice Hall, 2003).

15 For an expansion on the notion of Habits see, for example, Charles Camic, "The Matter of Habit," *American Journal of Sociology* 91 (1986): 1039–87.



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design and craft, which has followed the lead of product design and taken a research turn. The object or goal varies. Some doctoral studies have built products; others have built services, organizations, and programs. The corpus builds on a constructive approach; we have not included studies that have no practical roots.

Our use of the term *corpus* builds on the understanding articulated by Annie Gentes. She defines *corpus* as a group of texts that belong to the same genre. They are similar in style, narrative structure, and visual organization, and target the same audiences with similar textual strategies. Ultimately, the experience of the audience decides what belongs to a corpus.[16]

The main body of our corpus consists of 65 well-referenced and influential dissertations. We selected PhD level work for analysis because almost every writer in our corpus has a design background, and every single dissertation describes at least one design experiment and uses it as a vehicle for creating knowledge. In writing a doctoral thesis, design researchers have encountered methodological and theoretical questions coming from more established disciplines; they have found ways of reconciling these questions with their own beliefs, and they have had to describe these ways in detail. Our corpus reveals these struggles and shows how the researchers have found their way through them.

The PhD corpus has several desirable properties for us. Because it consists of PhD theses, it is much more explicit in terms of philosophical and theoretical orientation than design writing in general. The corpus is methodologically constructive; for example, we have excluded historical and philosophical theses from it.

Our method for analyzing the corpus is a combination of analytic induction[17] and explication. After creating a corpus of 65 theses, we divided it into primary and secondary categories according to how well we knew the theses. Our primary corpus consisted of 15 theses that we classified into four categories. (See Table 1.) Next, we examined each category closely to characterize the main features of the underlying epistemic organization. This step required close theoretical and philosophical contextualization of the thesis through its references, main arguments, and research program. Subsequently, we tested this categorization against the secondary corpus, which consisted of 50 additional theses. (See Table 2.) The final step validated the classification by analyzing theses authored by researchers whose background was not in Northwestern Europe. This step tested the framework against methodological and cultural variation and led to revisions of terminology but did not challenge the foundation laid in the previous steps. This process generated the categories we discuss in the next four sections and populated it with cases.

16 Annie Gentes, *The In-Discipline of Design* (Dordrecht: Springer, 2017), 90–91.

17 Ilpo Koskinen, "Empathic Design in Methodic Terms," in *Empathic Design*, ed. Ilpo Koskinen, Katja Battarbee, and Tuuli Mattelmäki (Helsinki: IT Press, 2003), 59–66.



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**Table 1 | Primary Corpus by Country**

| Denmark | Finland |
|---|---|
| 1. Bang, Anne Louise. "Emotional Value of Applied Textiles." PhD diss., Kolding School of Design, 2011. | 1. Acharya, Karthikeya. "Opening the Electrome." PhD diss., Aalto University, Helsinki, 2016. |
| 2. Dindler, Christian. "Fictional Space in Participatory Design of Engaging Interactive Environments." PhD diss., Aarhus University, 2010. | 2. Kurvinen, Esko. "Prototyping Social Action." PhD diss., University of Art and Design, Helsinki. 2007. |
| 3. Fogtmann, Maiken Hillerup. "Designing with the Body in Mind: Kinesthetic, Empathy, Interaction." PhD diss., Arkitektskolen Aarhus, 2011. | 3. Lee, Jung-Joo. "Against Method. The Portability of Method in Human-Centered Design." PhD diss., Aalto University, Helsinki, 2012. |
| 4. Kinch, Sofie. "Designing for Atmospheric Experiences—taking an Architectural Approach to Interaction Design." PhD diss., Aarhus School of Architecture, 2014. | 4. Mäkelä, Maarit. "Saveen piirtyvia muistoja." (in Finnish) ["Memories Written in Clay."] PhD diss., Aalto University, Helsinki, 2003. |
| 5. Krag, Mo Michelsen Stochholm. "Transformation on Abandonment: A New Critical Practice?" PhD diss., Aarhus School of Architecture, 2017. | 5. Mattelmäki, Tuuli. "Design Probes." PhD diss., University of Art and Design, Helsinki, 2006. |
| 6. Ludvigsen, Martin. "Designing for Social Interaction: Physical, Co-Located Social Computing." PhD diss., Aarhus School of Architecture, 2006. | 6. Summatavet, Kärt. "Folk Tradition and Artistic Inspiration." PhD diss., University of Art and Design, Helsinki, 2005. |
| 7. Lynggaard, Aviaja Borup. "Homing Interactions:-Tactics and Concepts for Highly Mobile People." PhD diss., Aarhus School of Architecture, 2012. | 7. Wu, Yiying. "Bicycles and Plants." PhD diss., Aalto University, Helsinki, 2017. |
| 8. Rasmussen, Majken Kirkegaard. "Changing the Shape of Interaction: Shape-Changing Interfaces." PhD diss., Aarhus University, 2015. | |

### Methodic Tradition

One tradition sees knowledge as an outcome of methodic rigor. A drift from the original starting point is justified if the chosen methods show that something in the initial assumptions has been wrong. Any other source of drifting is seen as method variance, which must be controlled. Drifting is a threat to rationality and the rational worldview behind this vision: Any act of drifting needs to be grounded in the production of methodic knowledge, and any act of drifting should be justified by this knowledge. The methodic tradition has a long history in design. The best-known examples are the design methods movement exemplified by John Christopher Jones's *Design Methods* and by Lucienne Blessing and Amaresh Chakrabarti.[18]

The fundamental tenets of the methodic tradition have been formulated particularly well by Gaver, who described its basic beliefs in *The Presence Project*:

> In pursuing its goal of producing causal explanations of the world, what distinguishes science from similar endeavors (e.g., religion) is that the methods employed to establish facts, construct theories, and relate the two

18 John Chris Jones, *Design Methods* (New York: John Wiley & Sons, 1992); and Lucienne Blessing and Amaresh Chakrabarti, DRM, a *Design Research Methodology* (London: Springer-Verlag, 2009).



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**Table 2 | Secondary Corpus by Country**

| Finland | Other Countries |
|---|---|
| 1. Aakko, Maarit. "Fashion in Between." PhD diss., Aalto University, Helsinki, 2016. | 1. Auger, James. "Why Robot? Speculative Design, the Domestication of Technology and the Considered Future." PhD diss., Royal College of Art, London, 2011. |
| 2. Ahde-Deal, Petra. "Women and Jewelry." PhD diss., Aalto University, Helsinki. 2013 | 2. Bakker, Saskia. "Design for Peripheral Interaction." PhD diss., Technische Universiteit Eindhoven, 2013. |
| 3. Battarbee, Katja. "Co-Experience: Understanding User Experiences in Social Interaction." PhD diss., University of Art and Design, Helsinki, 2004. | 3. Bruns, Miguel. "Relax: Inherent Feedback During Product Interactions to Reduce Stress." PhD diss., Technische Universiteit Eindhoven, 2010. |
| 4. Berg, Arild. "Artistic Research in Public Space." Aalto University, Helsinki, 2014. | 4. Busch, Otto von. "Fashion-Able: Hacktivism and Engaged Fashion Design." PhD diss., University of Gothenburg, 2008. |
| 5. Botero, Andrea. "Expanding Design Spaces. Design in Communal Endeavors." PhD diss., Aalto University, Helsinki, 2013. | 5. Deckers, Eva. "Perceptive Qualities in Systems of Interactive Products." PhD diss., Technische Universiteit Eindhoven, 2013. |
| 6. Garcia, Claudia Garduno. "Design as Freedom." PhD diss., Aalto University, Helsinki. 2017 | 6. Djajaginingrat, Tom. "Cubby: What You See is Where You Act." PhD diss., Delft University of Technology, 1998. |
| 7. Grondahl, Laura. "Experiences in Theatrical Space." PhD diss., University of Art and Design, Helsinki, 2004. | 7. Dunne, Anthony. "Hertzian Tales: Electronic Products, Aesthetic Experience, and Critical Design." PhD diss., Royal College of Art, London, 1999. |
| 8. Groth, Camilla. "Making Sense through Hands." PhD diss., Aalto University, Helsinki, 2017. | 8. Frens, Joep. "Designing for Rich Interaction: Integrating Form, Interaction, and Function." PhD diss., Technische Universiteit Eindhoven, 2006. |
| 9. Isohanni, Tuula. Arabia, "*Arabia: taiteellinen toiminta osana asuinympäristön suunnittelua, tapaus Arabianranta, Helsinki*" (in Finnish), [Arabia, Arabia: Artistic Activity in Planning a Housing District, Case Arabianranta, Helsinki.] PhD diss., University of Art and Design, Helsinki, 2006. | 9. Hengeveld, Bart. "Designing LinguaBytes. A Tangible Language Learning System for Non- or Hardly Speaking Toddlers." PhD diss., Technische Universiteit Eindhoven, 2011. |
| 10. Jacucci, Giulio. "Interaction as Performance." University of Oulu, 2004. | 10. Hummels, Caroline. "Gestural Design Tools: Prototypes, Experiments and Scenarios." PhD diss., Delft University of Technology, 2000. |
| 11. Jucide, Andrea. "Design for Hope." PhD diss., Aalto University, Helsinki, 2014. | 11. Keller, Ianus. "For Inspiration Only." PhD diss., Delft University of Technology, 2005. |
| 12. Judice, Marcelo. "You Are Important!" PhD diss., Aalto University, Helsinki, 2014. | 12. Kuusk, Kristi. "Crafting Sustainable Smart Textile Services." PhD diss., Technische Universiteit Eindhoven, 2016. |
| 13. Kabiito, Richard. "Meaning-Making in Visual Culture: The Case of Integrating Ganda Indigenous Knowledge with Contemporary Art Practice in Uganda." PhD diss., Aalto University, Helsinki. 2014 | 13. Niedderer, Kristina. "Designing the Performative Object: A Study in Designing Mindful Interaction through Artefacts." PhD diss., University of Plymouth, 2004. |
| 14. Kohtala, Cindy. "Making Sustainability." PhD diss., Aalto University, Helsinki, 2016. | 14. Raijmakers, Bas. "Design Documentaries. Using Documentary Film to Inspire Design." PhD diss., Royal College of Art, London, 2007. |
| 15. Kymalainen, Tiina. "Science Fiction Prototypes as Design Outcome of Research." PhD diss., Aalto University, Helsinki, 2015. | 15. Ross, Philip. "Ethics and Aesthetics in Intelligent Product and System Design." PhD diss., Technische Universiteit Eindhoven, 2008. |
| 16. Lankoski, Petri. "Character-driven Game Design." PhD diss., Aalto University, Helsinki, 2010. | 16. Sleeswijk-Visser, Froukje. "Bringing the Everyday Life of People into Design." PhD diss., Delft University of Technology, 2009. |
| 17. Lucero, Andres. "Co-designing Interactive Spaces with and for Designers." PhD diss., Technische Universiteit Eindhoven, 2009. | 17. Tromp, Nymke. "Social Design." PhD diss., Delft University of Technology, 2013. |
| 18. Marttila, Tatu. "Platforms of Co-Creation." PhD diss., Aalto University, Helsinki, 2018. | |

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**Table 2** | **Secondary Corpus by Country** (continued from previous page)

| Finland | Other Countries |
|---|---|
| 19. Miettinen, Satu. "Designing the Creative Tourism Experience." PhD diss., University of Art and Design, Helsinki, 2007. | 18. Trotto, Ambra. "Rights through Making: Skills for Pervasive Ethics." PhD diss., Technische Universiteit Eindhoven, 2011. |
| 20. Nimkulrat, Nithikul. "Paperness: Expressive Material in Textile Art from an Artist's Viewpoint." PhD diss., University of Art and Design, Helsinki, 2009. | 19. Vines, John. "Ageing Futures: Towards Cognitively Inclusive Digital Media Productions." PhD diss., University of Plymouth, 2011. |
| 21. Paavilainen, Heidi. "Dwelling with Design." PhD diss., Aalto University, Helsinki. 2013. | 20. Wallace, Jayne. "Emotionally Charged: A Practice-Centred Enquiry of Digital Jewellery and Personal Emotional Significance." PhD diss., Shefield-Hallam University, 2007. |
| 22. Saad-Sulonen, Johanna. "Combining Participations. Expanding the Locus of Participatory E-Planning By Combining Participatory Approaches in the Design of Digital Technology and in Urban Planning." PhD diss., Aalto University, Helsinki, 2014. | 21. Wensveen, Stephan. "A Tangibility Approach to Affective Interaction." PhD diss., Technische Universiteit Eindhoven, 2004. |
| 23. Säde, Simo 2001. "Cardboard Mock-ups and Conversations." PhD diss., University of Art and Design, Helsinki, 2001. | 22. Worbin, Linda. "Designing Dynamic Textile Patterns." PhD diss., University of Borås, 2010. |
| 24. Salgado, Mariana. "Designing for an Open Museum." PhD diss., University of Art and Design, Helsinki, 2009. | |
| 25. Scheible, Jürgen. "Empowering Mobile Art Practice." PhD diss., Aalto University, Helsinki, 2010. | |
| 26. Siikamäki, Raija. "Glass Can Be Recycled Forever Utilisation of End-of-Life Cathode Ray Tube Glasses in Ceramic and Glass Industry." PhD diss., University of Art and Design, Helsinki, 2006. | |
| 27. Soini, Katja. "Toward Resident-oriented Housing Renovation with Collaborative Design." PhD diss., Aalto University, Helsinki, 2015. | |
| 28. Vaajakallio, Kirsikka. "Design Games as a Tool, a Mindset and a Structure." PhD diss., Aalto University, Helsinki, 2012. | |
| 29. Ylirisku, Salu. "Frame It Simple!" PhD diss., Aalto University, Helsinki, 2013. | |

> are used to justify the validity of one's work, not [to justify] the results of these methods. No matter how plausible or desirable a statement or theory, it does not count as science unless the basis for asserting it is clear and veritable. Conversely, no matter how implausible or even dangerous a statement ("the earth revolves around the sun"), it is a scientific fact if the means used to assert it are sound.[19]

When we turn to our corpus, the methodic epistemic tradition is particularly strong in several Dutch dissertations from the technical universities in Delft and Eindhoven. The production is rich and the dissertation often true to the tradition. For example, "Designing for Rich Interaction" was the title of the dissertation of Joep Frens, as previously noted.[20] Building on J. J. Gibson's

19 *Presence Project* (London: RCA), 202–03.

20 Frens, "Designing for Rich Interaction: Integrating Form, Interaction, and Function," 176–85.



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ecological psychology, Frens believes that interaction design should take into account the form of a product, but only if designers pay attention to the user's cognitive, perceptual-motor, and emotional skills. On the basis of his research interest and potential contribution—the rich interaction framework—Frens focused on a tangible digital camera as the designed product to be tested; the camera became the pivot point of Frens's thesis work to learn about the suggested framework. The selection of design alternatives and choices were rationally argued, which turned the cameras into hypotheses that could be tested.

Drifts that follow the methodic tradition have undeniable merits. When methods are clear and everyone follows them, inspecting a work to see whether it is correct is easy. In addition, if everything is correct but something still goes against expectations, researchers know they are on to something. By basing knowledge claims on methodic rigor, a research community forms an intersubjective community that can locate its problems quickly and produce results that can be defended against demagogues who contest the validity of scientific knowledge, as Gaver implies in the quote above.

**Experiential Tradition**

Identifying the science-inspired tradition rendered what we term an "experiential" tradition in the background—often craft-based studies that find their correlates from art and design rather than the sciences or the humanities, and where the ultimate arbiter of knowledge is a designer and her experience. In craft research, authors like Kristina Niedderer have argued in favor of accepting practical experience as a source of knowledge.[21] In *The Presence Project*, Gaver and colleagues compared this practical experience with the demands of epistemological accountability: Because design is aesthetically accountable, if a design concept works, whether the methods for building it are solid does not matter.[22] Designers do not need to justify their methods because designers produce objects, not explanations; and if the design works, its origin does not matter.

We saw two variants of this experiential tradition. The first variant gives a nod to the traditional studio. In her PhD work, Jayne Wallace articulated a way to give more freedom to design decisions that are hard to capture in rational terms.[23] For her, drifting took place as part of a studio-based practice. The value and credibility of the work depends primarily on her skills. Some other designers believe that not even studio practice is needed to justify drifting. Kärt Summatavet's thesis, "Folk Tradition and Artistic Inspiration," had a constructive part that was shaped by a folkloristic study.[24]

21 Kristina Niedderer and Seymour Roworth-Stokes, "The Role and Use of Creative Practice in Research and Its Contribution to Knowledge," in *Proceedings of IASDR, Hong Kong SAR2007* (Hong Kong: The Hong Kong Polytechnic University, 2007),

22 *Presence Project*, 202–03.

23 Jayne Wallace, "Emotionally Charged: A Practice-Centred Enquiry of Digital Jewellery and Personal Emotional Significance" (Sheffield: Sheffield-Hallam University, 2007).

24 Kärt Summatavet, "Folk Tradition and Artistic Inspiration: A Woman's Life in Traditional Estonian Jewelry and Crafts as Told by Anne and Roosi" (Phd diss., University of Art and Design Helsinki, 2005).



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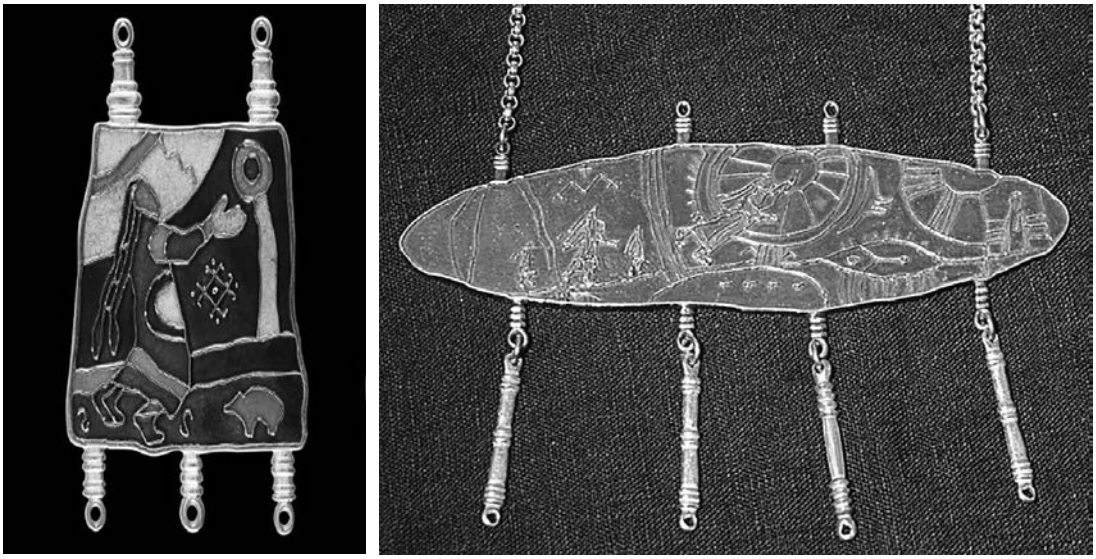


Figure 1
(left) The Maiden and the Moon, gold and enamel, by Kart Summatavet (2001).
(right) The Setu Bride Leaves Her Home – Joys, silver, by Kart Summatavet (2001).

She built three exhibitions to explore jewelry in three stages of a woman's life: childhood, maturity, and old age. The objects were exquisitely crafted; assessed by professional goldsmiths; exhibited publicly in high-end museums in Helsinki and Tallinn; and evaluated by researchers, designers, and artists.

Summatavet consistently separated her field studies and artwork and refused to talk about the connection even in her public examinations. This approach was accepted in the design school context in which she was working (see Figure 1).

The main validity criterion was the connection between her fieldwork and her artwork. Summatavet clearly had a sound rationale for this criterion. Following her ethnographic-inspired research of the folkloristic tradition was easy, and its main conclusions provided the foundation for her three exhibitions. Research gave her the design path. For girls, Estonian traditional goldsmiths used copper and a specific set of traditional symbols. The set was different from the fertile age when the preferred metal was silver. It was also different from the post-menopausal years when the metal was gold. Summatavet's research provided her exhibitions with a storyline, and researchers assessed this storyline. The details in her enamel work were left for the artistic judgment. Although she refused to reveal her process, her craft was exquisite, and the main dimensions of her symbolic universe in the exhibitions came from her ethnographic research. Her stance was respected because the logic behind it was clear and the artworks achieved a very high professional level. There was no need to justify the path from inspiration to art works with a method.

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**Drifting as Dialectic**

The discovery and description of the experiential tradition led us to conduct another round of sequencing. In this process, we realized a third way of building knowledge—one that is at the core of participatory and empathic design. In several dissertations from Aarhus and Helsinki, users were actively involved in the research. In this case, drifting research becomes a function of knowledge produced in the interaction between users and designers. It is a function of a dialectic between researchers and users.

For example, in the early days of participatory design (PD), the goal of participatory processes was to develop technology and systems design that would accomplish three things: *Enable quality* of the technologies, *emancipate* the workers, and *democratize* the organization of work. One of the main insights of participatory design built on a lesson that Pelle Ehn and his fellow researchers learned from graphic workers they studied in their *UTOPIA project* in the 1970s.[25] They realized that they must use mock-ups and other low-level prototypes to create meaningful communication with prospective users. Earlier attempts to use flowcharts and other scientific tools proved to be futile. Users pushed the change; researchers responded, often against their scientific inclinations. Researchers were successful when they gave up the belief that their knowledge was superior to that of the involved users.

To illustrate, Christian Dindler sees his research in participatory terms.[26] His research delivers insights on how institutions, such as museums and archives, may move their conception of exhibition concepts from the perspective of visitors as passive recipients of experiences to active participants who co-construct engaging experiences. He does so by narrating what he terms "Fictional Spaces," in which participants may co-imagine acts and connections of the exhibited artifacts. Thus, the objective of the design activity is not predominantly to create an artifact, but to engender a capacity to involve participants and enable them potentially to change their perception of what may be an engaging museum.

Dindler's academic legacy influenced another project—one in which among others Peter Gall Krogh and Ole Sejer Iversen also participated.[27] A regional museum of archaeology in Denmark approached us with two concerns: 1) The number of visitors to the museum was decreasing; and 2) because of Danish treasure regulations, their most attractive artifact—a mummified bronze-age woman—had been transferred to the National Museum. As a remedy, the museum applied for and was granted funds to install a digital (i.e., virtual reality (VR)) version of the woman. However, in talks with the museum staff, the designers learned that the archeological process was of greater interest to the public and the staff than the actual artifacts exhibited. Based on this perspective, the

25 Pelle Ehn, "Work-Oriented Design of Computer Artifacts" (Stockholm: Arbetslivscentrum, 1988).

26 Christian Dindler, "Fictional Space in Participatory Design of Engaging Interactive Environments" (PhD diss., Aarhus University, 2010).

27 Christian Dindler et al., "Engagement Through Mixed Modalities," *Interactions* 18, no. 4 (2011): 34–39.



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Figure 2
An interactive burial mound. Digital images are projected into the dirt and revealed through removal of dirt. Photo by Christian Dindler (2010).

project drifted from implementing a VR installation to conceptualizing and installing a digitally enhanced sand pitch that allowed visitors to repeatedly excavate items and invite archeological reflections (see Figure 2). Furthermore, following the shift, the institution decided to change its institutional identity: Rather than a "museum," it was a "center of archeology."

By carefully listening to stories of both staff and involved museumgoers, Dindler, fellow designers, and museum staff drifted from the design intention of simulating a specific historic artifact to reconceptualizing what it means to unpack and tell stories of the past. This sort of user-driven (re)specification is at the very heart of the dialectic tradition, which sees knowledge in the context of the community rather than in the context of a method or the design experience. In the dialectic tradition, users' knowledge must be respected, even if it runs against researchers' scientific and artistic inclinations and the clients' assumptions. Although researchers need to be wary of the biases and power structures of the community with which they work the foundation of the tradition is the dialectic between users' and researchers' knowledge structures.

**Programmatic Tradition**

Yet another tradition locates knowledge in discourse. It has found support especially in design-oriented HCI.[28] As research debates evolve, they become increasingly informed, and drift occurs. This evolution and drift slowly improve the frameworks researchers use and introduces discourse as potential contribution. The name *programmatic tradition* reflects the way that research programs are both affirmed and challenged. In this tradition, researchers pay attention

28 John Zimmerman et al., "Research Through Design as a Method for Interaction Design Research in HCI," in *Proceedings of Conference on Human Factors in Computing Systems* (New York: ACM, 2007), 493–502.



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to the conceptual and theoretical scaffoldings around design artifacts. These artifacts are indexical and open to many interpretations. As researchers create frameworks and theories and debate their pros and cons, the meaning of the design artifacts keeps changing. Drifting, in this view, may happen in design projects themselves, as well as in their conceptual environment. The growth of knowledge results from improvements in understanding, rather than from new empirical facts or design artifacts.

The doctoral study of Helsinki-based designer and professor Tuuli Mattelmäki and its pivotal character exemplify this epistemological tradition.[29] Mattelmäki's early experiments with probes followed in the footsteps of cultural probes—inspired by and critical towards how those probes privileged designers over users.[30] And unlike the London group led by Bill Gaver and Anthony Dunne, she worked in an industrial context that demanded evidence. By 2002, she had added an interpretive validation step to the probing process, and she published her work slightly later.[31] Her work also has been subject to methodological and historical reinterpretations.[32]

From a broader perspective, Mattelmäki's work shows how revisions of earlier research can lead to multiple outcomes in a research program.[33] Her work gets its meaning from the program, and its meaning drifts as time goes by and her work is discussed by others. In the programmatic tradition, knowledge is doubly contextual: it elaborates on earlier research, but its meaning is also dependent on future developments and evolutions.

As Redström has noted, the programmatic tradition shifts attention to the dialectic between design and its theoretical background.[34] The vantage point of any knowledge claim is coherence in the research community, but this goalpost is always shifting. The conversation between research and theory is typical to the natural sciences, as well as to the humanities and the social sciences.[35] It also is familiar to design, where fashion and even architecture similarly progress mostly through reinterpretations rather than through new discoveries.

The programmatic tradition may sound counterintuitive in that it sees research artifacts not as self-standing statements but as negotiations between many sources of knowledge.[36] However, knowing this tradition may help to resolve some of the differences between the other traditions. For example, a study can be methodologically perfect, but even the best studies can be surpassed by later research. Or the intuitions of designers change in the experiential tradition: What is good today may not be good 20 years from now in a dramatically different context. Studio practice does progress. Designers like Summatavet and Wallace build on references and precedents, and their work paves a way for subsequent

29 Tuuli Mattelmäki, "Design Probes" (PhD diss., Aalto University, 2006).

30 Bill Gaver et al., "Design: Cultural Probes," *Interactions* 6, no. 1 (1999): 21–29.

31 Tuuli Mattelmäki, "Probes: Studying Experiences for Design Empathy," in *Empathic Design*, ed. Ilpo Koskinen et al., (Helsinki: IT Press, 2003), 119–30.

32 Tuuli Mattelmäki et al., "What Happened to Empathic Design," *Design Issues* 30, no. 1 (Winter 2014): 67–77, https://doi.org/10.1162/DESI_a_00249.

33 Anne Louise Bang and Mette Agger Eriksen, "Experiments All the Way in Programmatic Design Research," *Artifact: Journal of Design Practice* 6, no. 1–2 (2014): 8.1–8.20.

34 Johan Redström, "Some Notes on Program-Experiment Dialectics," in *Proceedings of Nordes '11* (Helsinki: Nordes, 2011), https://archive.nordes.org/index.php/n13/index (accessed March 3, 2021).

35 For the former, see Imre Lakatos, "Falsification and the Methodology of Scientific Research Programmes," in *Criticism and the Growth of Knowledge* 4, ed. Imre Lakatos and Alan Musgrave (Cambridge: Cambridge University Press, 1970), 91–196; and P. Feyerabend, *Against Method: Outline of an Anarchist Theory of Knowledge* (New York: Verso Books, 1975). For the latter, see Clifford Geertz, *The Interpretation of Cultures* (New York: Basic Books, 1973), 25–26.

36 Cf. Schmid, "The Philosophical Underpinnings of Design Theory," 415–30.



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research. The dialectical tradition also progresses: Events and interpretations perceived as normal and correct do change over time. Theoretical discourse creates reinterpretations and recontextualizations that change the meaning of the work—sometimes years later. Perhaps the best case is the meaning of the UTOPIA project—one of the foundations of participatory design. Ehn originally argued, as unpacked in his doctoral dissertation (cited earlier), that the success of Participatory Design could be explained by the fact that it created a Wittgensteinian language game that joined users and designers. However, twenty years later, his explanation of the project was Latourian: UTOPIA functioned like a parliament of things by bringing people together to a reasoned discussion.[37]

**Conclusions and Discussion**

This article has studied drifting from an epistemological perspective. It has explicated four concepts of knowledge from a corpus of 65 doctoral theses that mostly have their origin in Northern Europe. Our interpretation of epistemology owes more to laboratory studies and the sociology of scientific knowledge than to traditional philosophical epistemology. We see epistemology as a practice, and our goal has been to explicate this practice from within.

We have described four epistemological traditions operating within constructive design research. In the *methodic* tradition, designers follow a method. If the method tells them that their original beliefs were wrong, they must accept their defeat and adjust their interpretation accordingly, as Jones and as Blessing and Chakrabarti did. In contrast, in the *experiential* tradition, drifting is driven by the studio process, as in Wallace, or by the designer's intuition, as in Summatavet. In the *dialectic* tradition processes typical to participatory and empathic design, drift is justified when user studies show that the researchers were wrong. Finally, the *programmatic* tradition shifts the argument to discourse that broadens interpretive possibilities that enter the discourse wherever they find their meaning.[38]

If our analysis is valid, design researchers have at least four belief systems that influence how they justify knowledge. These different systems may seem to be incommensurate. Yet all of them have led to interesting and widely admired pieces of work. As we have pointed out, they have many similarities as well.

This article has contributed to constructive design research by focusing on its epistemological foundations. In regard to the validity of our analysis, the method of this article has been analytical induction, which has a well-known limit[39]: This process generates results that are valid in the corpus only. The corpus is predominantly produced in Northwestern Europe, although it

37 See Pelle Ehn, "Participation in Design Things," in Proceedings of Participatory Design Conference PDC'08 (2008), 92–101.

38 See Lakatos, "Falsification and the Methodology of Scientific Research Programmes" and Redström, *Making Design Theory*.

39 Clive Seale, *The Quality of Qualitative Data* (London: Sage, 1999).



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includes studies undertaken by designers from many countries and ethnic backgrounds. In addition, roughly half of the theses were written by female researchers. Design is a global occupation. We clearly need to avoid sweeping generalizations from one demographic and need to explore dissertations outside the current cultural scope. Beyond this corpus, future research might account, for example, for feminist epistemologies and for non-Western theoretical traditions.[40]

One of the philosophical inspirations of this article has been Anne-Françoise Schmid's deconstruction of epistemology in design.[41] She recently argued that the heterogeneity of the topics that science today must deal with makes framing all science within a single philosophical theory impossible. The same is true, she continues, in design, which in the construction of its object integrates many types of knowledge. Seen through Schmid's perspective, our epistemological interpretation is part of a larger shift toward contextual epistemology—in other words, a shift in attention to the circumstances and preconditions that precede knowledge statements. Our interpretation also is part of a shift to a contextual (Schmid would say sociological) respecification of epistemology—a shift that started with Thomas Kuhn, who described scientific progress in terms of scientific revolution initiated by accumulating anomalies that challenge paradigmatic status quo, and has expanded from physics to feminism, biological sciences, and economics over the last forty years.[42]

In terms of our ethics, we hope to expand tolerance in design research and to diminish practices and perspectives that police what is right and what is wrong. We have been fascinated by the fact that design researchers of various persuasions communicate without difficulties at a practical level, even though they can radically disagree about the very foundations of their activity, including the concept of knowledge. The disagreement most often may result from their conflicting interpretations of what constitutes and drives knowledge. When we do constructive design research, we need to be aware of its underlying epistemological traditions and cherish the fact that the concepts can differ radically. What works in Jayne Wallace's craft may not be productive in Simo Säde's study of user interfaces to forest tractors—and vice versa.[43] If our article is able to increase tolerance in the design research community, we have achieved one of our ethical goals.

We also want to give design researchers tools to participate in the language of other fields of knowledge. Most design researchers work in multidisciplinary teams, in which epistemic traditions may differ. Being aware of these differences is a hallmark of a discipline whose researchers can collaborate with people whose interpretations of knowledge can differ radically.

40 See, e.g., Maarit Mäkelä, Saveen piirtyviä muistoja [Memories Written in Clay] (Helsinki: Aalto University, 2003); and Andrea Judice, "Design for Hope."

41 Schmid, "The Philosophical Underpinnings," 415–30.

42 Thomas Kuhn and Ian Hacking, *The Structure of Scientific Revolutions: 50th Anniversary Edition* (Chicago: University of Chicago Press, 2012); Sandra Harding, Whose Science? Whose Knowledge? (Ithaca, N.Y.: Cornell University Press, 1991); Alex Rosenberg and Daniel W. McShea, *Philosophy of Biology: A Contemporary Introduction* (New York and London: Routledge, 2008); Uskali Mäki, Realism and Economic Methodology (London: Routledge, 2007); Harold Kincaid and Don Ross (eds.), *The Oxford Handbook of Philosophy of Economics* (New York: Oxford University Press, 2009).

43 Simo Säde, "Cardboard Mock-ups and Conversations" (Phd diss., University of Art and Design, 2001).

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